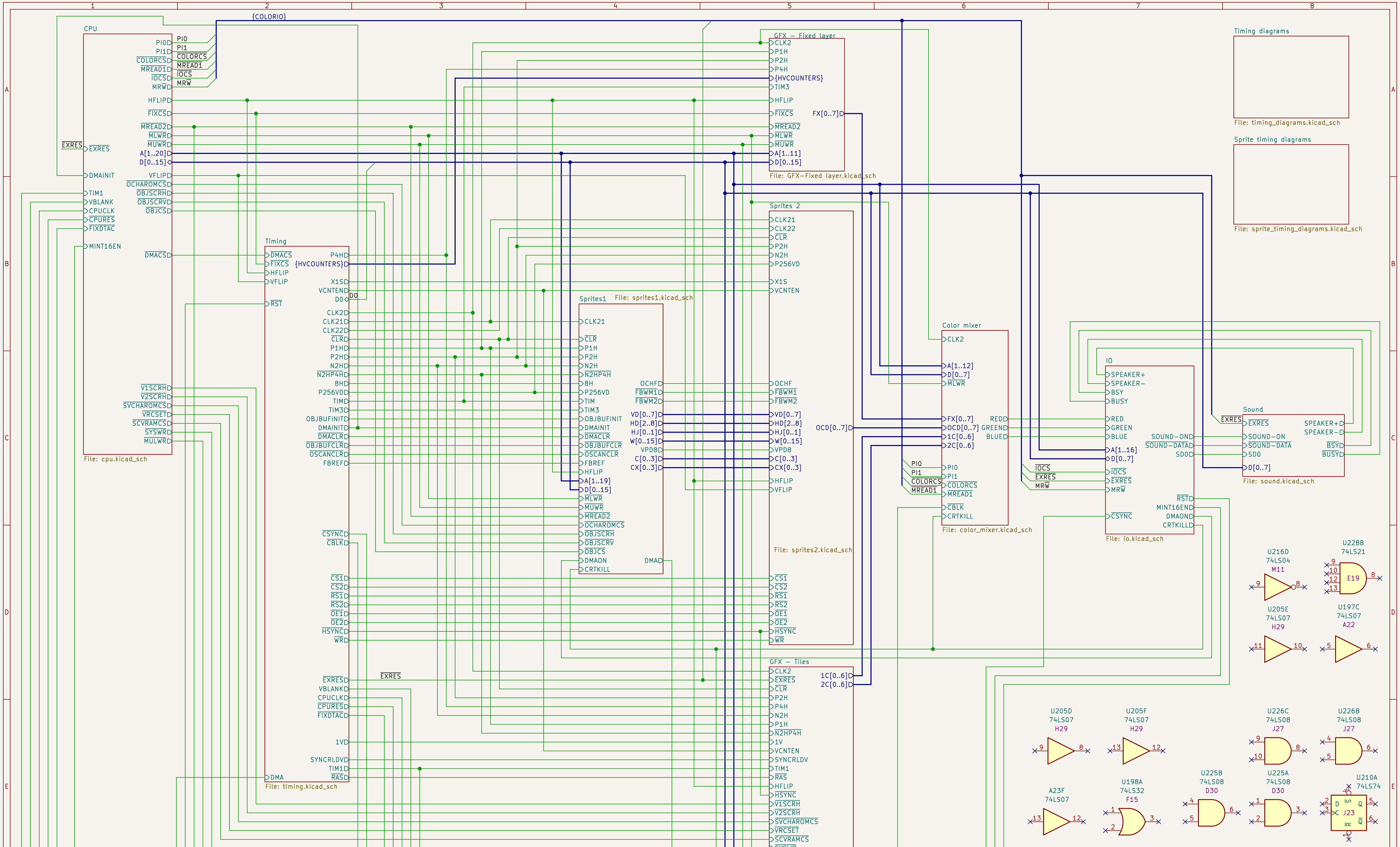


Ulf Skutnabba, twitter: @skutis77

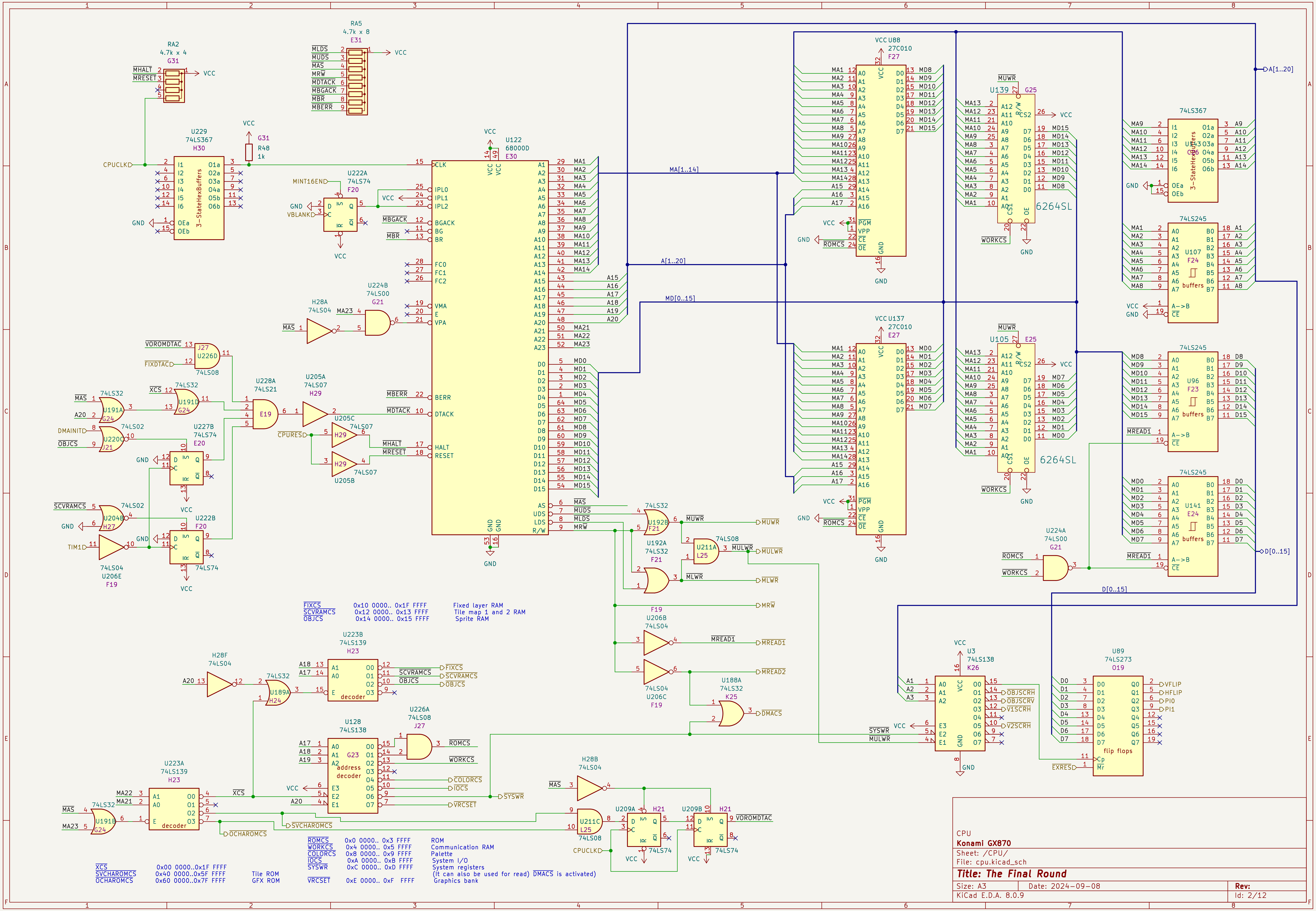
Konami GX870

Sheet: /

File: fround.kicad_sch

Title: The Final Round

Size: A3	Date: 2024-09-08	Rev:
KiCad E.D.A. 8.0.9		Id: 1/12



FIXCS 0x10 0000.. 0x1F FFFF Fixed layer RAM
SCVRAMCS 0x12 0000.. 0x13 FFFF Tile map 1 and 2 RAM
OBJCS 0x14 0000.. 0x15 FFFF Sprite RAM

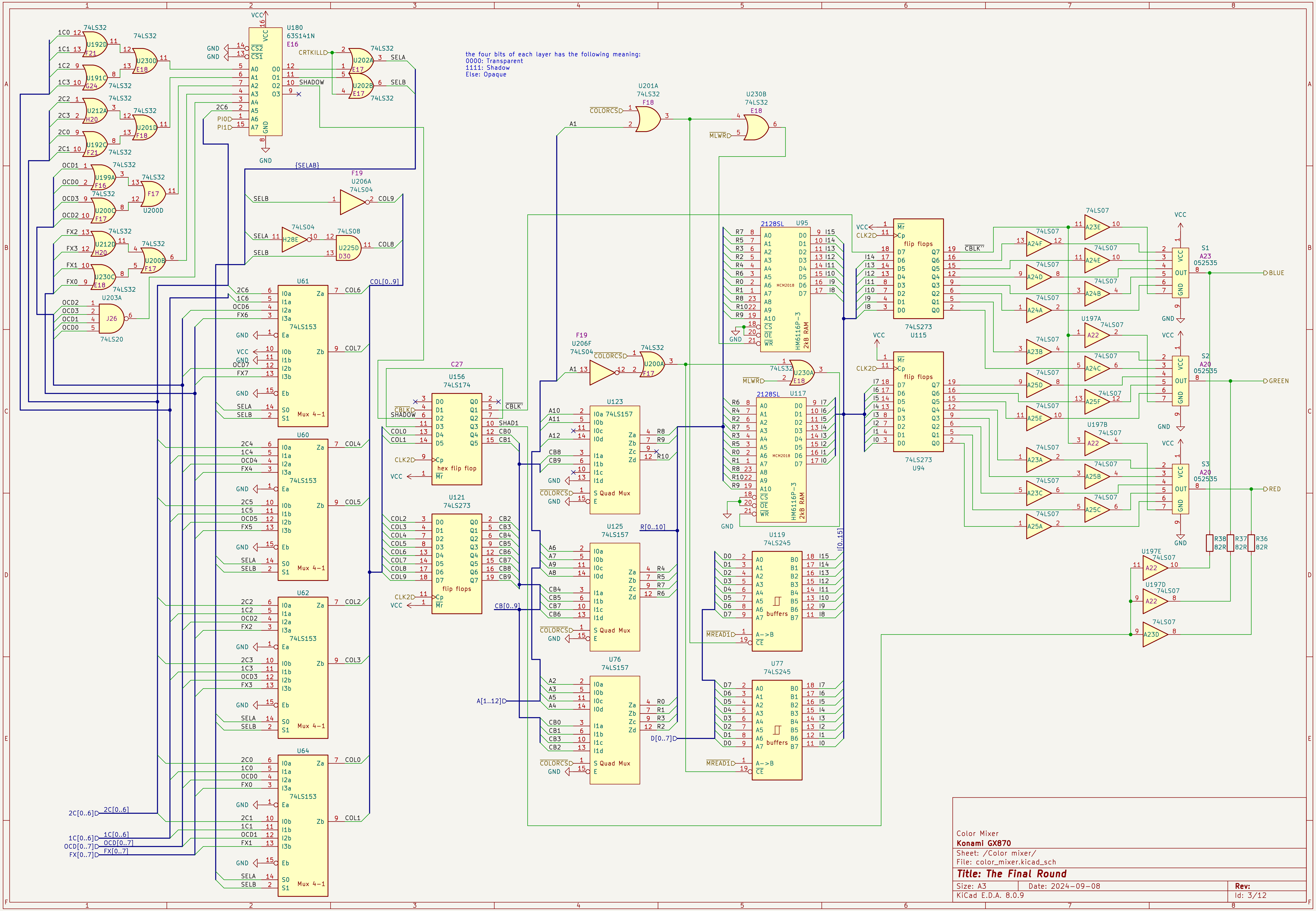
ROMCS 0x0 0000.. 0x3 FFFF ROM
WORKCS 0x4 0000.. 0x5 FFFF Communication RAM
COLORCS 0x8 0000.. 0x9 FFFF Palette
IOCS 0xA 0000.. 0xB FFFF System I/O
SYSWR 0xC 0000.. 0xD FFFF System registers
(It can also be used for read) DMACS is activated)
VRCSET 0xE 0000.. 0xF FFFF Graphics bank

XCS 0x00 0000..0x1F FFFF
SVCHAROMCS 0x40 0000..0x5F FFFF Tile ROM
OCHAROMCS 0x60 0000..0x7F FFFF GFX ROM

CPU
Konami GX870
Sheet: /CPU/
File: cpu.kicad_sch

Title: The Final Round

Size: A3 Date: 2024-09-08 Rev:
KiCad E.D.A. 8.0.9 Id: 2/12

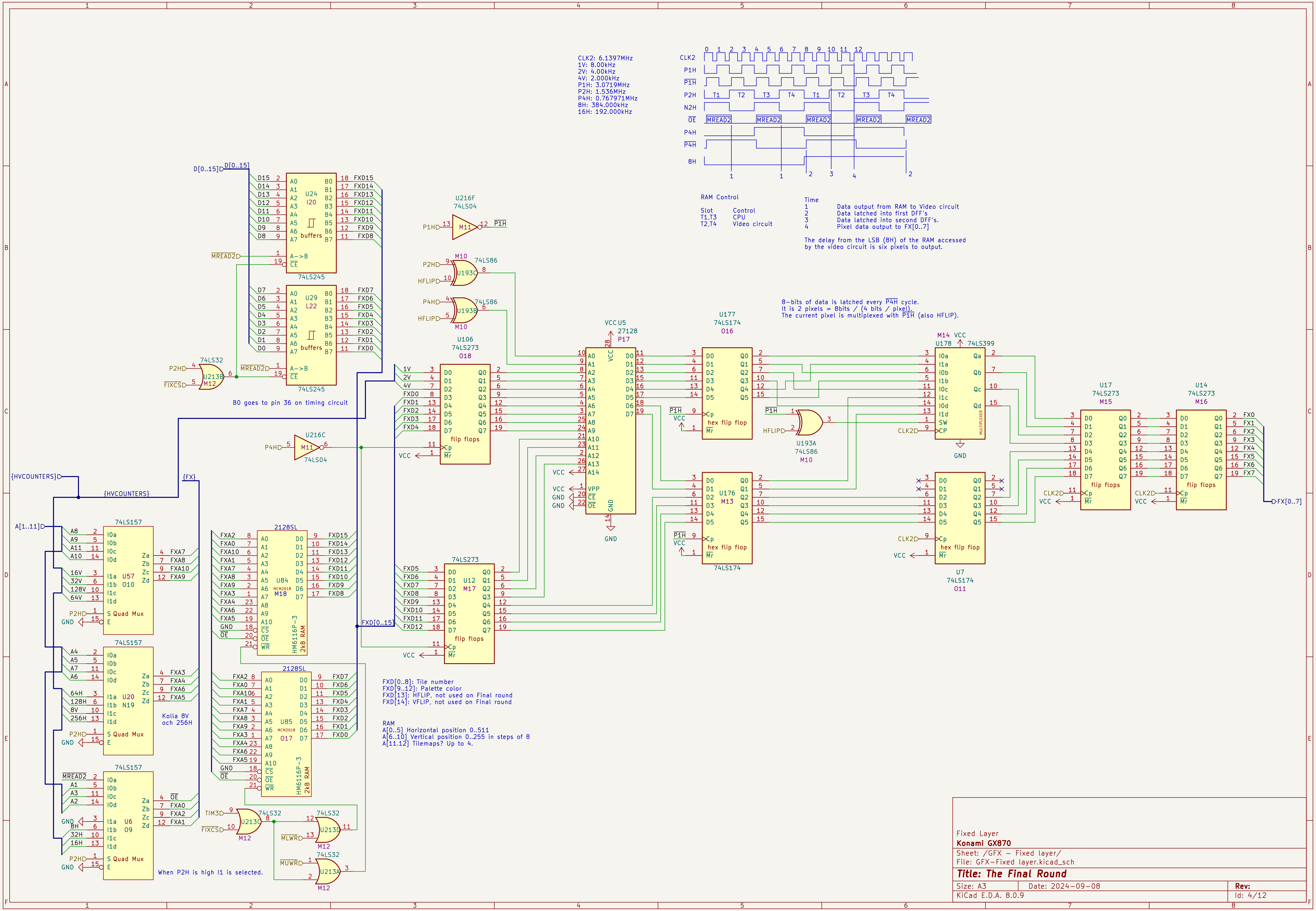


the four bits of each layer has the following meaning:
0000: Transparent
1111: Shadow
Else: Opaque

Color Mixer
Konami GX870
Sheet: /Color mixer/
File: color_mixer.kicad_sch

Title: The Final Round

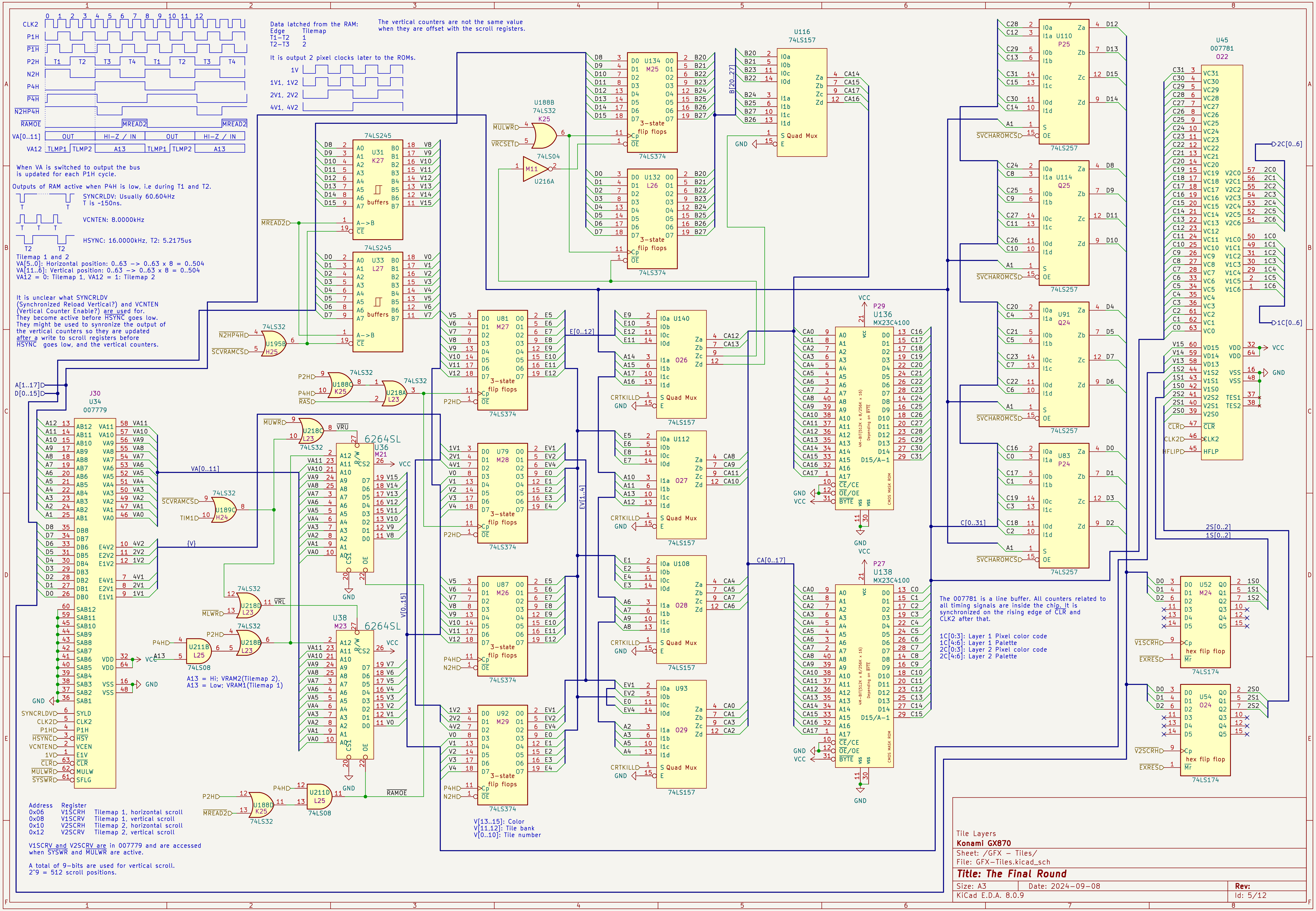
Size: A3 Date: 2024-09-08 Rev:
KiCad E.D.A. 8.0.9 Id: 3/12

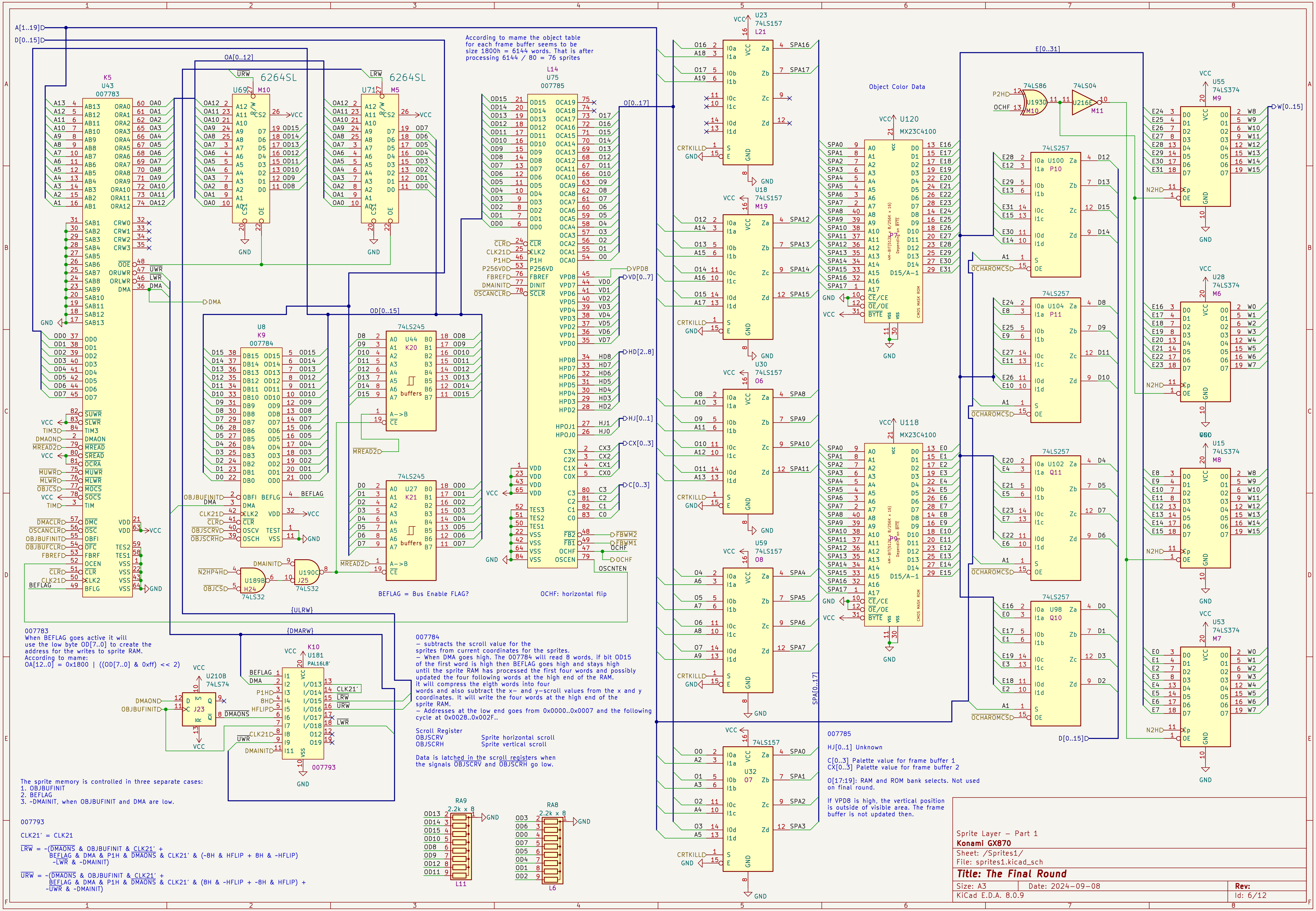


Fixed Layer
Konami GX870
Sheet: /GFX - Fixed layer/
File: GFX-Fixed layer.kicad_sch

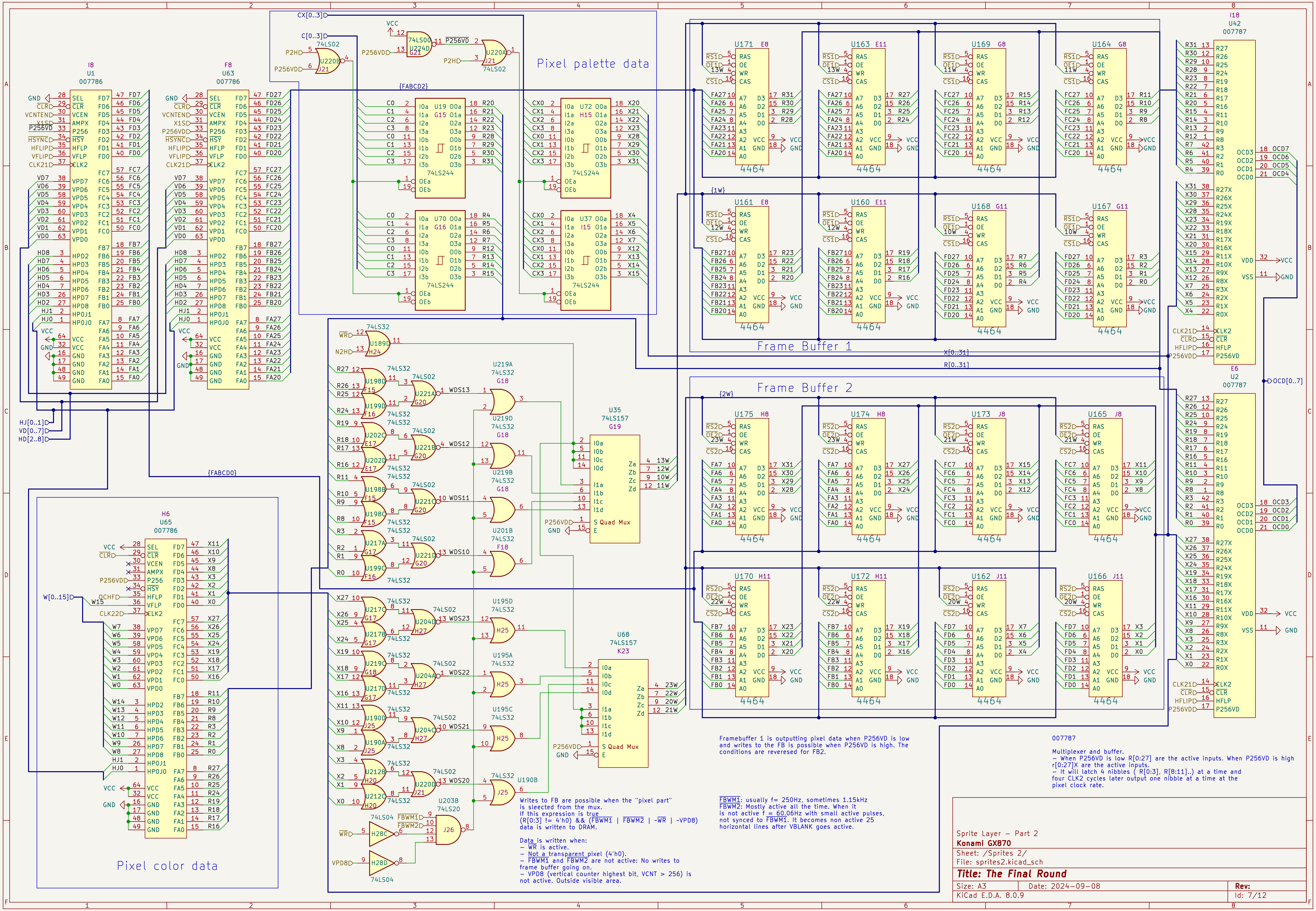
Title: The Final Round

Size: A3 | Date: 2024-09-08 | Rev:
KiCad E.D.A. 8.0.9 | Id: 4/12

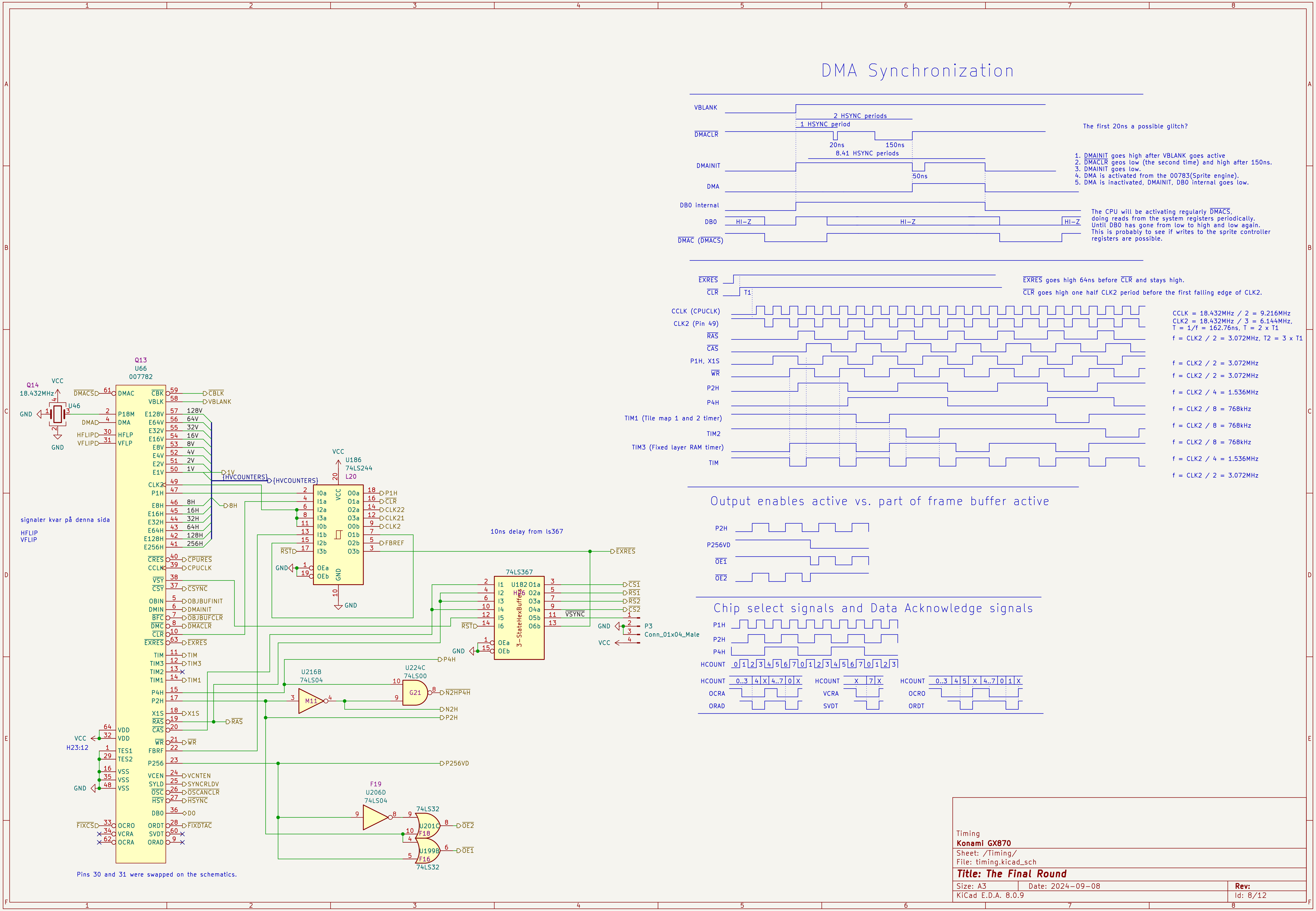

Data latched from the RAM:
Edge Tilemap
T1–T2 1
T2–T3 2
It is output 2 pixel clocks later to the ROMs.
The vertical counters are not the same value when they are offset with the scroll registers.
When VA is switched to output the bus is updated for each P1H cycle.
Outputs of RAM active when P4H is low, i.e during T1 and T2.
SYNCRLDV: Usually 60.604Hz
T is ~150ns.
VCNTEN: 8.0000kHz
HSYNC: 16.0000kHz, T2: 5.2175us
Tilemap 1 and 2
VA[5..0]: Horizontal position: 0..63 -> 0..63 x 8 = 0..504
VA[11..6]: Vertical position: 0..63 -> 0..63 x 8 = 0..504
VA12 = 0: Tilemap 1, VA12 = 1: Tilemap 2
It is unclear what SYNCRLDV (Synchronized Reload Vertical?) and VCNTEN (Vertical Counter Enable?) are used for. They become active before HSYNC goes low. They might be used to synronize the output of the vertical counters so they are updated after a write to scroll registers before HSYNC goes low, and the vertical counters.
A13 = Hi: VRAM2(Tilemap 2).
A13 = Low: VRAM1(Tilemap 1)
V[13..15]: Color
V[11,12]: Tile bank
V[0..10]: Tile number
Address Register
0x06 V1SCRH Tilemap 1, horizontal scroll
0x08 V1SCRV Tilemap 1, vertical scroll
0x10 V2SCRH Tilemap 2, horizontal scroll
0x12 V2SCRV Tilemap 2, vertical scroll
V1SCRV and V2SCRV are in 007779 and are accessed when SYSWR and MULWR are active.
A total of 9-bits are used for vertical scroll.
2^9 = 512 scroll positions.
The 007781 is a line buffer. All counters related to all timing signals are inside the chip. It is synchronized on the rising edge of CLR and CLK2 after that.
1C[0:3]: Layer 1 Pixel color code
1C[4:6]: Layer 1 Palette
2C[0:3]: Layer 2 Pixel color code
2C[4:6]: Layer 2 Palette
Tile Layers
Konami GX870
Sheet: /GFX – Tiles/
File: GFX-Tiles.kicad_sch
Title: The Final Round
Size: A3 Date: 2024-09-08 Rev:
KiCad E.D.A. 8.0.9 Id: 5/12



According to mame the object table for each frame buffer seems to be size 1800h = 6144 words. That is after processing 6144 / 80 = 76 sprites
Object Color Data
BEFLAG = Bus Enable FLAG?
OCHF: horizontal flip
007783
When BEFLAG goes active it will use the low byte OD[7..0] to create the address for the writes to sprite RAM.
According to mame:
OA[12..0] = 0x1800 | ((OD[7..0] & 0xff) << 2)
007784
– subtracts the scroll value for the sprites from current coordinates for the sprites.
– When DMA goes high. The 007784 will read 8 words, if bit OD15 of the first word is high then BEFLAG goes high and stays high until the sprite RAM has processed the first four words and possibly updated the four following words at the high end of the RAM.
it will compress the eigth words into four words and also subtract the x– and y–scroll values from the x and y coordinates. It will write the four words at the high end of the sprite RAM.
– Addresses at the low end goes from 0x0000..0x0007 and the following cycle at 0x0028..0x002F..
Scroll Register
OBJSCRV Sprite horizontal scroll
OBJSCRH Sprite vertical scroll
Data is latched in the scroll registers when the signals OBJSCRV and OBJSCRH go low.
007785
HJ[0..1] Unknown
C[0..3] Palette value for frame buffer 1
CX[0..3] Palette value for frame buffer 2
O[17:19]: RAM and ROM bank selects. Not used on final round.
If VPD8 is high, the vertical position is outside of visible area. The frame buffer is not updated then.
The sprite memory is controlled in three separate cases:
1. OBJBUFINIT
2. BEFLAG
3. ~DMAINIT, when OBJBUFINIT and DMA are low.
007793
CLK21′ = CLK21
LRW = ~(DMAONS & OBJBUFINIT & CLK21′ +
BEFLAG & DMA & P1H & DMAONS & CLK21′ & (~8H & HFLIP + 8H & ~HFLIP)
~LWR & ~DMAINIT)
URW = ~(DMAONS & OBJBUFINIT & CLK21′ +
BEFLAG & DMA & P1H & DMAONS & CLK21′ & (8H & ~HFLIP + ~8H & HFLIP)
~UWR & ~DMAINIT)
Sprite Layer – Part 1
Konami GX870
Sheet: /Sprites1/
File: sprites1.kicad_sch
Title: The Final Round
Size: A3
Date: 2024-09-08
Rev:
KiCad E.D.A. 8.0.9
Id: 6/12



Pixel palette data
Frame Buffer 1
Frame Buffer 2
Pixel color data
U1 007786
U63 007786
U65 007786
U42 007787
U2 007787
U19 74LS244
U72 74LS244
U70 74LS244
U37 74LS244
U35 74LS157 Quad Mux
U68 74LS157 Quad Mux
U171 4464
U163 4464
U169 4464
U164 4464
U161 4464
U160 4464
U168 4464
U167 4464
U175 4464
U174 4464
U173 4464
U165 4464
U170 4464
U172 4464
U162 4464
U166 4464
Framebuffer 1 is outputting pixel data when P256VD is low and writes to the FB is possible when P256VD is high. The conditions are revereesed for FB2.
Writes to FB are possible when the "pixel part" is sleected from the mux.
If this expression is true
(R[0:3] != 4'h0) && (FBWM1 | FBWM2 | ~WR | ~VPD8)
data is written to DRAM.
Data is written when:
– WR is active.
– Not a transparent pixel (4'h0).
– FBWM1 and FBWM2 are not active: No writes to frame buffer going on.
– VPD8 (vertical counter highest bit, VCNT > 256) is not active. Outside visible area.
FBWM1: usually f= 250Hz, sometimes 1.15kHz
FBWM2: Mostly active all the time. When it is not active f = 60.06Hz with small active pulses, not synced to FBWM1. It becomes non active 25 horizontal lines after VBLANK goes active.
007787
Multiplexer and buffer.
– When P256VD is low R[0:27] are the active inputs. When P256VD is high r[0:27]X are the active inputs.
– It will latch 4 nibbles (R[0:3], R[8:11]..) at a time and four CLK2 cycles later output one nibble at a time at the pixel clock rate.
Sprite Layer – Part 2
Konami GX870
Sheet: /Sprites 2/
File: sprites2.kicad_sch
Title: The Final Round
Size: A3
Date: 2024-09-08
Rev:
KiCad E.D.A. 8.0.9
Id: 7/12



Timing

Konami GX870

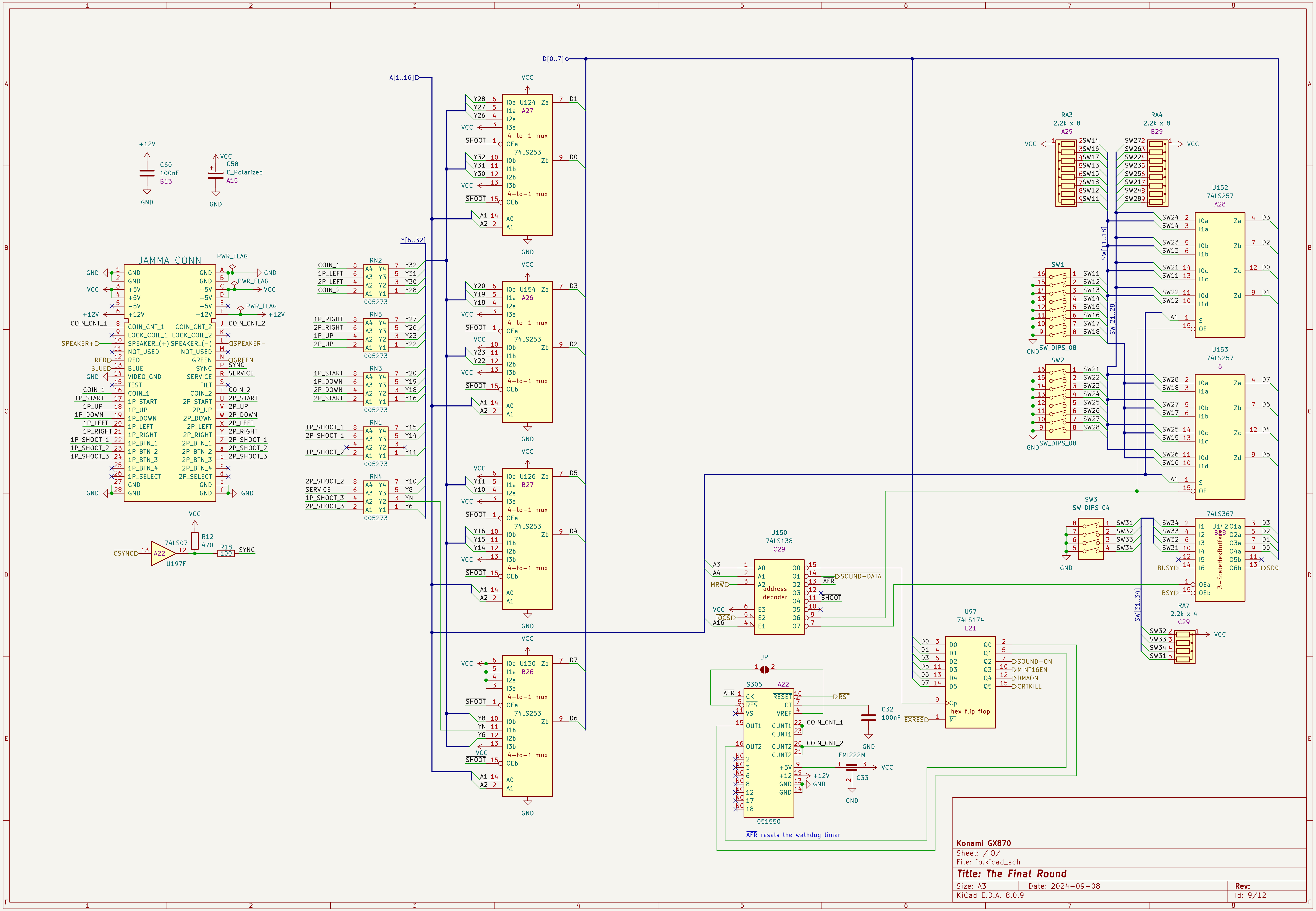
Sheet: /Timing/

File: timing.kicad_sch

Title: The Final Round

Size: A3 Date: 2024-09-08 Rev:

KiCad E.D.A. 8.0.9 Id: 8/12



Konami GX870

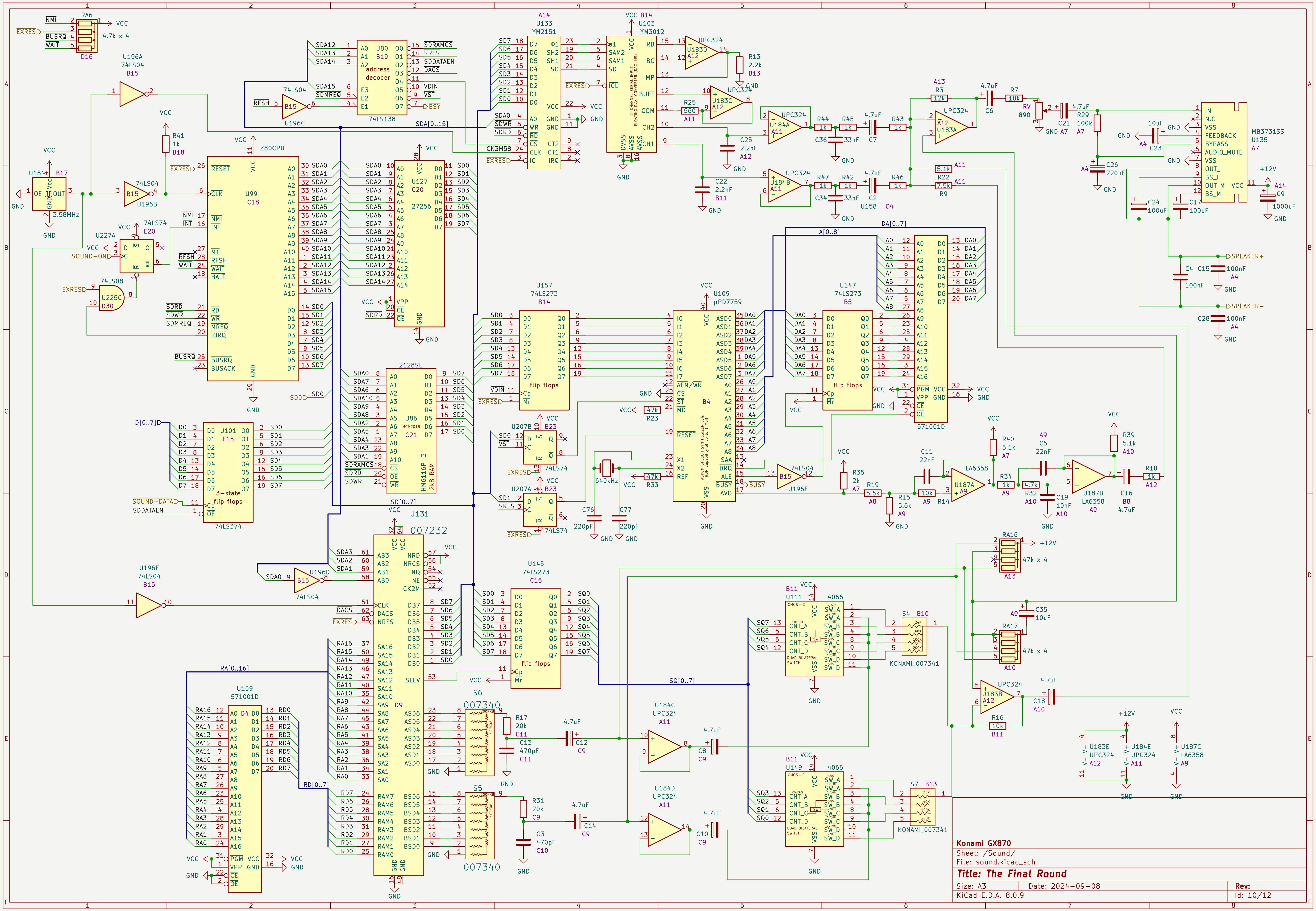
Sheet: /IO/

File: io.kicad_sch

Title: The Final Round

Size: A3 Date: 2024-09-08 Rev:

KiCad E.D.A. 8.0.9 Id: 9/12



Konami GX870

Sheet: /Sound/

File: sound.kicad_sch

Title: The Final Round

Size: A3 | Date: 2024-09-08 | Rev:

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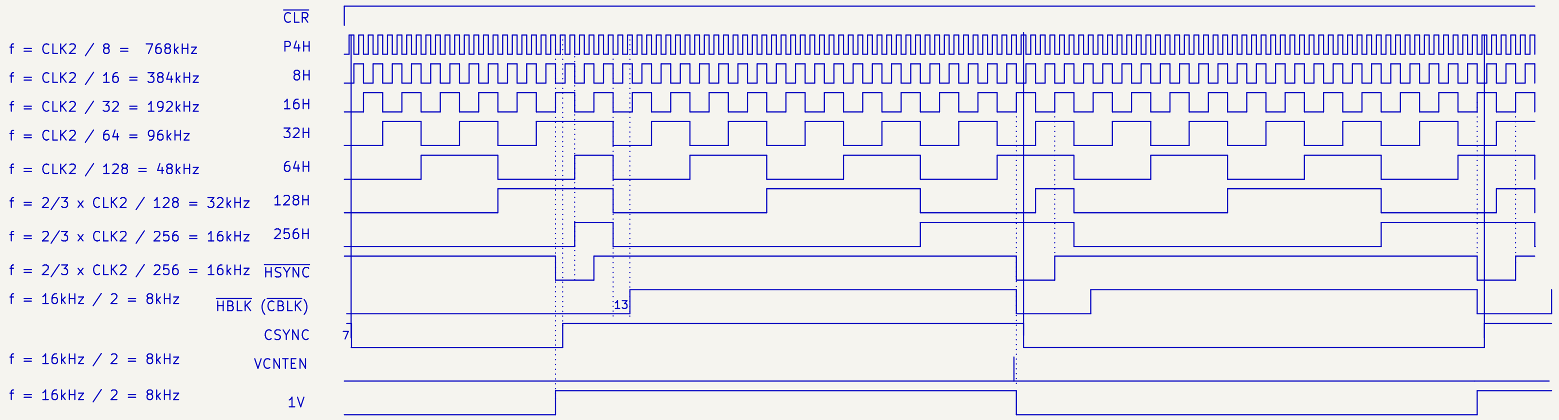
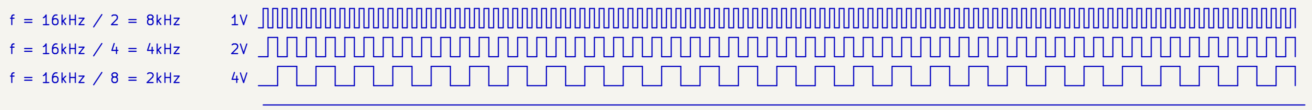
Horizontal signals

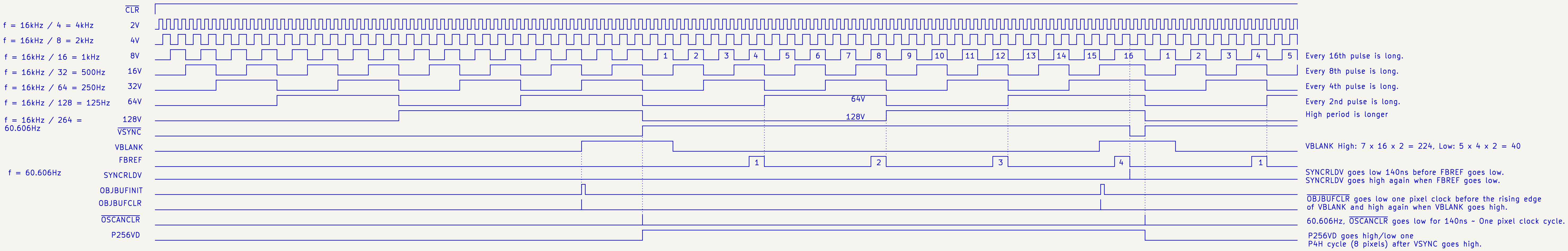


- The first VCNTEN is skipped after reset. It goes low 140ns before HSYNC goes low, and high again when HSYNC goes low. VCNTEN is active right before every second falling edge of HSYNC.
- CPURES goes high, and stays high, on the seventh falling edge of HSYNC.

The first HSYNC is at HCOUNT 175 after reset.

Vertical signals





FBREF is measured at pin 22. At U186:3 FBREF is delayed by 22ns going high and 12ns going low.

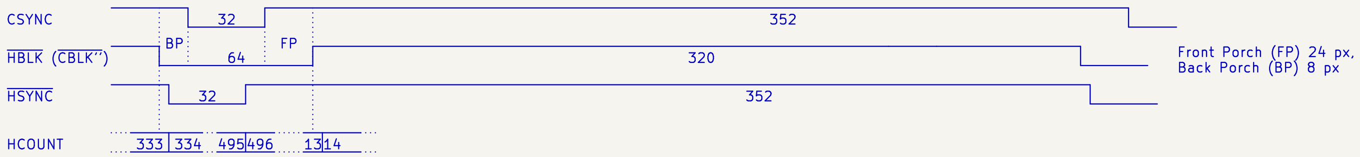
Horizontal and vertical synch timing diagrams

Pixel clock CLK2 = 18.432MHz / 3 = 6.144MHz

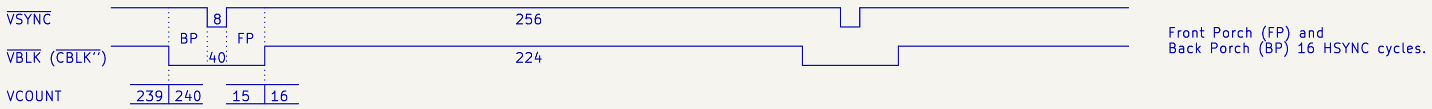
Horizontal lines
HSYNC freq = 6.144MHz / 384 = 16kHz
HSYNC period = 1 / 16kHz = 62.5us
Pixels / line: 384
Active pixels: 320
Blank pixels: 64

Vertical lines
VSYNC freq = 16kHz / 264 = 60.606060Hz
VSYNC period = 1 / f = 264 / 16kHz = 16.5ms
scanlines / frame: 264
Active lines: 224
Blank lines: 40

The numbers in the HSYNC and HBLK diagrams are HSYNC cycles.
All edges are synchronised to the rising edge of CLK2.



The numbers in the VSYNC and VBLK diagrams are HSYNC cycles.
All edges are synchronised to the falling edge of HSYNC.



HCOUNT is bits [256H, 128H, 64H, 32H, 16H, 8H, P4H, P2H, P1H]
VCOUNT is bits [128V, 64V, 32V, 16V, 8V, 4V, 2V, 1V]

CBLK'' is at the output of color mixer.

Konami GX870
Sheet: /Timing diagrams/
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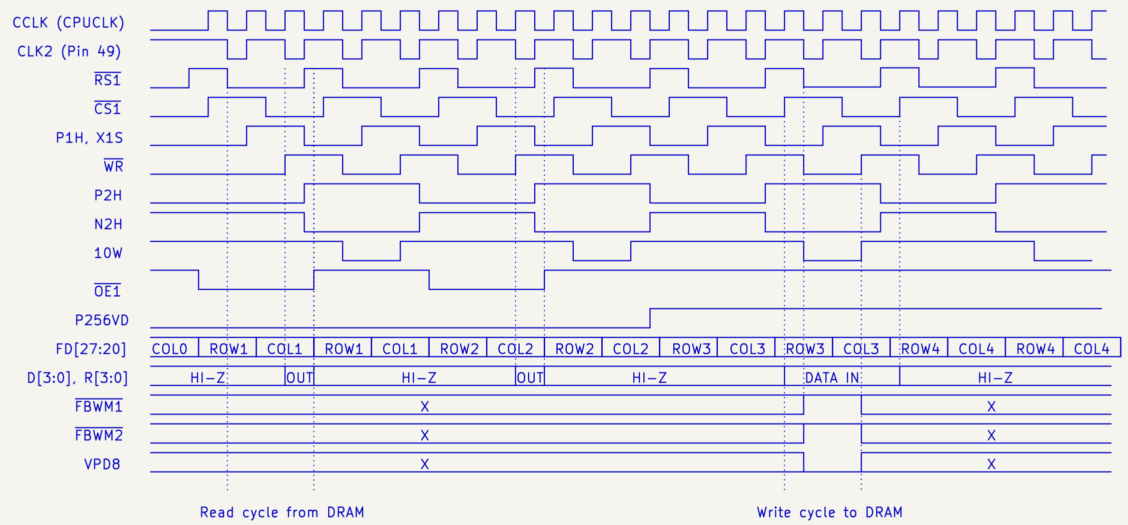

The same row and column addresses are repeated twice.
One for read cycle and the other for write cycles.

Konami GX870
Sheet: /Sprite timing diagrams/
File: sprite_timing_diagrams.kicad_sch

Title: The Final Round

Size: A3 | Date: 2024-09-08 | **Rev:**
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